

Car Seat Sales Forecasting Using Multi-Linear Regression

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Executive Summary

This analysis applies a **Multi-Linear Regression (MLR)** model to forecast car seat sales across retail stores using store-level demographic, pricing, and merchandising attributes.

MLR serves as a **transparent, interpretable baseline** that allows direct estimation of the influence of each predictor on sales. The model is trained using the historical dataset and evaluated using **cross-validated performance metrics** to ensure comparability with other models in this project.

The trained model is applied to **new store data** to generate forward-looking sales forecasts supporting:

- Inventory and demand planning
- Store-level pricing strategy
- Advertising budget optimization
- Risk mitigation for underperforming store configurations

This MLR model complements CART, Random Forest, and Neural Network approaches, offering **interpretability and actionable insights** into sales drivers.

1. Project Overview

Multi-Linear Regression is a **classic, interpretable regression technique** that models the relationship between multiple predictors and a continuous outcome.

In this notebook: - **Target variable:** Sales

- **Predictors:** Store-level pricing, advertising, demographics, merchandising
 - **Evaluation:** 5-fold cross-validation for performance benchmarking
 - **Prediction:** Applied to new store-level data
-

2. Business Objective

The MLR model aims to: - Forecast car seat sales across retail stores

- Quantify the influence of individual predictors
 - Provide a transparent baseline model for comparison
 - Generate predictions for new stores for operational decisions
-

3. Dataset Overview

The dataset contains the following variables:

Target Variable

- **Sales:** Number of car seats sold (in hundreds)

Predictor Variables

- **CompPrice** – Competitor pricing
- **Income** – Local average income (in \$000s)
- **Advertising** – Advertising spend (in \$000s)
- **Population** – Local population size (in \$000s)
- **Price** – Store price
- **ShelfLoc** – Shelf location quality (Good, Medium, Bad)
- **Age** – Average customer age
- **Education** – Average education level
- **Urban** – Urban location indicator
- **US** – U.S. store indicator

Excluded Variable

- **Store** – Unique store identifier
-

4. Environment Setup

```
library(caret)
library(dplyr)
library(ggplot2)
```

5. Data Ingestion

```
car_seat_sales <- read.csv("data/car_seat_sales.csv")
new_car_seat_sales <- read.csv("data/new_car_seat_sales_data.csv")
```

6. Data Preparation

The `Store` variable is removed as it is a non-predictive identifier.

Categorical variables are explicitly converted to factors and aligned between training and new data.

```
library(dplyr)

model_data <- car_seat_sales %>%
  select(-Store)

new_model_data <- new_car_seat_sales %>%
  select(-Store)

categorical_vars <- c("ShelfLoc", "Urban", "US")

for (col in categorical_vars) {
  model_data[[col]] <- as.factor(model_data[[col]])
  new_model_data[[col]] <- factor(
    new_model_data[[col]],
    levels = levels(model_data[[col]])
  )
}
```

7. Modeling Strategy

Why Multi-Linear Regression?

Multiple Linear Regression provides a transparent and interpretable framework for understanding the relationship between sales and multiple predictors.

Key advantages include:

- Clear interpretation of predictor effects
- Strong baseline for comparison with non-linear models
- Useful for explaining sales drivers to business stakeholders

Validation Approach

A 5-fold cross-validation strategy is used to:

- Assess model generalization
- Produce performance metrics (RMSE, R^2)
- Enable fair comparison with CART, Random Forest, and Neural Networks

8. Model Training – Multi-Linear Regression

```

library(caret)

set.seed(1975)

cv_method <- trainControl(
  method = "cv",
  number = 5
)

mlr_model <- train(
  Sales ~ .,
  data = model_data,
  method = "lm",
  trControl = cv_method
)

mlr_model

## Linear Regression
##
## 400 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 321, 319, 321, 320, 319
## Resampling results:
##
##   RMSE      Rsquared   MAE
##  5.232661  0.0100576  4.479052
##
## Tuning parameter 'intercept' was held constant at a value of TRUE

```

9. Model Evaluation

```

summary(mlr_model$finalModel)

##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.4982 -4.0172 -0.2135  4.1992  9.6804
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.6802887   3.0354037   2.860  0.00447 **
## CompPrice     0.0043383   0.0208631   0.208  0.83538
## Income      -0.0113825   0.0092811  -1.226  0.22078
## Advertising  -0.0170072   0.0559532  -0.304  0.76133
## Population    0.0004475   0.0018634   0.240  0.81035

```

```
## Price          0.0070010  0.0134358   0.521  0.60261
## ShelfLocGood  -0.7502538  0.7701576  -0.974  0.33059
## ShelfLocMedium -1.1167261  0.6343233  -1.760  0.07911 .
## Age           -0.0114908  0.0160043  -0.718  0.47320
## Education      0.0077817  0.0991959   0.078  0.93751
## UrbanYes       0.4360498  0.5682804   0.767  0.44336
## USYes          0.1327926  0.7537209   0.176  0.86024
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.125 on 388 degrees of freedom
## Multiple R-squared:  0.01748,    Adjusted R-squared:  -0.01038
## F-statistic: 0.6275 on 11 and 388 DF,  p-value: 0.8055
```

Interpretation guidelines:

- Coefficients indicate the expected change in sales per unit change in predictors
- R^2 measures the proportion of variance explained
- RMSE represents prediction error magnitude

10. Generating Predictions on New Store Data

```
mlr_predictions <- predict(mlr_model, new_model_data)

prediction_results <- new_car_seat_sales %>%
  mutate(Predicted_Sales = mlr_predictions)

head(prediction_results)
```

```
##   Store CompPrice Income Advertising Population Price ShelfLoc Age Education
## 1   216      116     83          15         170   144     Bad   71         11
## 2    8      136     81          15         425   120     Good   67         10
## 3   273      113     33           0          14    63     Good   38         12
##   Urban  US Predicted_Sales
## 1   Yes  Yes      8.906485
## 2   Yes  Yes      8.250029
## 3   Yes  No      8.584749
```

`Predicted_Sales` represents the expected number of car seats sold, measured in **hundreds of units**, for each new store.

11. Business Insights

Multiple Linear Regression enables clear identification of factors influencing sales:

- **ShelfLoc** and **Price** exhibit the strongest impact on sales
- Demographic variables such as **Age**, **Income**, and **Education** show smaller but measurable effects
- The linear baseline helps validate and calibrate more complex models

12. Limitations and Next Steps

Limitations of the Multi-Linear Regression model include:

- Assumption of linear relationships between predictors and sales
- Sensitivity to multicollinearity among predictors
- Inability to capture complex non-linear interactions

Planned next steps:

- Regularized regression (Ridge, Lasso, Elastic Net)
- Cross-model performance comparison
- Final model selection for deployment